The Boundary is a Rate: Coherence–Rupture–Regeneration and the Unevidenceable Boundary

Self-measurement, retensing and the collective now under the current CRR formulation

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Abstract. Sandved-Smith, Fields, Doctor, Laukkonen and Hohwy argue that a persisting, self-evidencing system can never obtain evidence that it is separate from its environment, and read the realisation of emptiness as the embodied recognition of this impossibility. I examine their boundary through Coherence–Rupture–Regeneration (CRR).

In CRR:

- a system accumulates Fisher–Rao arc on a statistical carrier;
- it is cut at the first passage of a half-turn of net travel;
- it regenerates from an age-weighted mean of completed occasions.

Four results follow.

1. **Self-measurement overshoot.** The entropy a finite agent reads from its own state in any frame it can deploy exceeds the true von Neumann entropy, except in the eigenbasis. This is a direct consequence of the Schur–Horn theorem.
2. **Retensing.** The entanglement entropy across the boundary, the quantity the impossibility theorem forbids the agent to measure, is bounded by an accumulated temporal quantity: $S_A(t) \leq S_A(0) + \int \sqrt{V}\, dC$, where C is the Fisher–Rao arc of the system's spectrum and V its varentropy. The constant is scale-dependent: below 1 for a qubit, so that $C \geq S_A$, and above 1 from four levels upward. Accumulated coherence bounds the entanglement entropy outright only for small systems, and the bound is crossed by an exact geodesic construction from d = 14.
3. **Regeneration.** It is a normalised, many-to-one average, and so a contraction, not an amplifier.
4. **The collective now.** Many coupled CRR rotors undergo the Kuramoto synchronisation transition, giving an exact criterion for a shared present.

All results were preregistered and confirmed on fresh random instances. Together they support a reading in which the boundary is a rate rather than a surface: the present has no content of its own, and the far side of the boundary is the settled past of other finite systems.

Keywords: self-evidencing; emptiness; quantum reference frames; Fisher information; varentropy; Schur–Horn; synchronisation; process ontology.

1. Introduction

To persist is to keep confirming a model of what one is and what surrounds one. In the free-energy principle, a system that endures self-evidences: it minimises the surprisal of its sensory states and so accumulates evidence for its own generative model.

The impossibility result. Sandved-Smith and colleagues [1] identify one belief this process can never reach, namely that the system is separate from its environment. Confirming separateness would require quantifying the system's entanglement with everything it is not, and no finite system can measure the entanglement entropy across its own boundary from inside. Their argument runs as follows:

- separability is not merely unconfirmed but structurally unevidenceable;
- the belief in separation acts as a structural prior over the agent's quantum-reference-frame (QRF) deployments, built into its measurements and never among the things measured;
- realising emptiness can then be read as the embodied recognition of this impossibility.

An operational companion [2] states the central claim in software terms: a finite agent can use a boundary for prediction without ever obtaining evidence that the boundary is real.

What this paper does. I take this result as correct and ask what remains once the boundary is read through a temporal framework. CRR describes persisting systems as accumulating distinguishable structure, reaching the limit of a regime, and reconstructing from their own history. The paper:

- states the framework (§2);
- derives and tests four formal results that bear on the impossibility argument (§§3–4);
- develops the ontological reading those results support (§5).

2. Framework

All definitions follow the current CRR core.

- **Carrier and coherence.** A system's state is a point on a statistical carrier with the Fisher–Rao metric. Coherence C is the accumulated Fisher–Rao arc of its path, in fixed units σ. The chord C* is the geodesic distance between an occasion's start and end, and the surplus $S = C - C^*$ is the travel in excess of the chord.
- **The cut.** On an identifiable circular carrier of circumference ℓ, a cut occurs at the first passage of net forward travel through a half-turn: $t_{(n+1)} = \inf\{ t > t_n : s(t)$

$- s(t_n) = a \}$, with $a = \ell/2$ and $\Omega = 1/a$. On a completed half-turn with backward travel B, $S = 2B$.

- **Regeneration.** The seed of the next occasion is an age-weighted Fréchet mean of completed contents, $m_{n+1} \in \operatorname{argmin}_y \Sigma_j q^{n-j} d(y, \Phi_j)^2$, which in flat coordinates is the convex recursion $m_{n+1} = q\, m_n + (1 - q)\, \Phi_n$. Only completed occasions contribute.
- **Physical laws.** Laws such as overdamped motion on the carrier are constitutive additions to the framework.
- **The cost of the path.** At fixed duration T, frictional dissipation over an occasion obeys $Q\, T \geq \zeta_{\min} \sigma^2 (C^* + S)^2$, so surplus has a thermodynamic price.
- **Ontology.** CRR is paired with a growing-block reading: actual history accumulates, the present is an empty boundary that carries conditions inherited from that history, and future content is not actual.

3. Results

3.1 Self-measurement overshoot

Proposition 1. For any state ρ and any orthonormal basis, the Shannon entropy of the measurement outcomes is at least the von Neumann entropy S(ρ). Equality holds exactly when the basis diagonalises ρ.

Proof. The outcome probabilities are the diagonal of ρ in the chosen basis. By the Schur–Horn theorem this diagonal is majorised by the spectrum of ρ, and Shannon entropy is Schur-concave. ∎

An agent reading its own boundary entropy from any frame it can deploy therefore overshoots, unless it already knows its eigenbasis. Identifying the eigenbasis requires the full tomography and stable external reference that a finite system lacks for its own boundary. The obstruction proved in general by the impossibility theorem thus has an elementary, frame-level counterpart.

3.2 Retensing: a temporal bound on the forbidden correlation

The impossibility theorem forbids measuring $S_A = ½\, I(A:E)$, the system's correlation with its co-present complement. CRR invites the question whether this spatial quantity is controlled by a temporal one, accumulated along the system's own past.

Proposition 2. Let p(t) be the spectrum of $\rho_A(t)$, C(t) its accumulated Fisher–Rao arc, and $V(p) = \operatorname{Var}_p(-\ln p)$ its varentropy. Then

$$S_A(t) \leq S_A(0) + \int_0^t \sqrt{V}\, dC,$$

with equality along entropy-gradient flow.

Proof. Since $\Sigma\, dp_i = 0$, $dS = -\Sigma (\ln p_i + S)\, dp_i$. Cauchy–Schwarz with weights p_i gives $|dS| \leq \sqrt{V} \cdot \sqrt{(\Sigma\, dp_i^2/p_i)} = \sqrt{V}\, ds$. Integrate. ∎

The size of the constant. Let $\kappa(d)$ be the maximum of $\sqrt{V}$ over spectra on d outcomes, computed numerically:

d	2	4	8	32
$\kappa(d)$ (nats)	0.663	1.012	1.333	1.949

Three consequences follow:

- **Qubits.** For a qubit, $\kappa < 1$, so $C \geq S_A$ holds for every trajectory: accumulated coherence bounds the entanglement entropy outright.
- **Larger systems.** From $d = 4$ upward, $\kappa > 1$, and the outright bound can fail.
- **A geometric criterion.** The Fisher–Rao geodesic from a pure spectrum to the uniform one has length $2 \arccos(1/\sqrt{d})$. This exceeds the entropy it gains, $\ln d$, for $d \leq 13$, and falls short of it for $d \geq 14$. So beyond 13 levels a spectrum can gain more entropy than the arc it travels.

3.3 Regeneration is a contraction

Regeneration is a normalised, age-weighted average of bounded contents. It therefore:

- remains within their convex hull;
- maps many distinct records to the same seed.

It cannot amplify, and it is not invertible. Physically realising it means overwriting distinctions in a record, at a Landauer-type cost whose size depends on how the record is coarse-grained. Together with the dissipation bound of §2, this states precisely what persistence through regeneration costs: the surplus travelled within each occasion, plus the memory that compresses completed occasions into a seed.

3.4 The collective now

The model. Take N CRR rotors, each a phase s_i on its own circular carrier, with the overdamped law and mean-field coupling:

$\dot{s}_i = \tau_i + K r \sin(\psi - s_i)$, $r e^{i\psi} = \langle e^{i s_j} \rangle$.

This is exactly the Kuramoto model. With tilts τ_i drawn from a Lorentzian of half-width γ, there is a critical coupling $K_c = 2\gamma$:

- **above it**, $r = \sqrt{1 - K_c/K}$: the rotors share a common phase;
- **below it**, $r = O(N^{-1/2})$: only private phases survive in the large-N limit.

This gives a dynamical criterion for when many finite systems share a present.

4. Numerical confirmation

4.1 Design

The propositions above are exact. To check the implementation, and the scale at which the outright bound $C \geq S_A$ fails in physical dynamics, I preregistered eight predictions and confirmed them on fresh random instances (study `BOUNDARY-RATE-CHECK-2026-10-27`).

Pipeline:

1. **Development.** Thresholds were fixed on development seeds 0–2.
2. **Freeze.** Code, scorer and predictions were frozen (FREEZE sha256 `6d1a7669…94b9`).
3. **Anchor.** The freeze was anchored by a pushed commit (`b29a1c8`) and an OpenTimestamps stamp.
4. **Holdout.** Only then were the confirmation instances drawn: random generator 2000, trajectory seeds 2000–2005 and Kuramoto seeds 2000–2002.
5. **Reproducibility.** Two reruns reproduced all outputs byte for byte.

Models:

- **Self-measurement.** 500 random states with Haar-random bases at each of d = 2, 4, 8 and 16.
- **Retensing.** Systems of 1, 2, 3 and 5 qubits coupled to baths of 4–5 qubits through GUE Hamiltonians, starting from product pure states; 6 seeds per size and 241 time points.
- **Collective now.** N = 4,000 rotors at K/Kc = 1.5, 2 and 3, and N = 250–4,000 at K = Kc/2.

4.2 Outcomes

All eight predictions held.

- **Self-measurement.** The overshoot was non-negative in every case. Mean overshoot was 0.25, 0.38, 0.44 and 0.47 nats at d = 2, 4, 8 and 16, and equality held in the eigenbasis to 10^{-15}.
- **Retensing:**
  - the bound of Proposition 2 held at every time point for every system size;
  - $C \geq S_A$ held throughout for $d_A = 2$ and 4;
  - it failed on every seed at $d_A = 8$, by 0.03–0.05 nats;
  - it failed on every seed at $d_A = 32$, by 0.65–0.69 nats, where S_A reached 1.38 C (Figure 1).
- **Collective now.** Above threshold the order parameter matched $\sqrt{(1 - Kc/K)}$ to within 0.006. Below threshold r decayed with N at a log–log slope of −0.38, toward the incoherent value of −½ as run length grows (Figure 2).

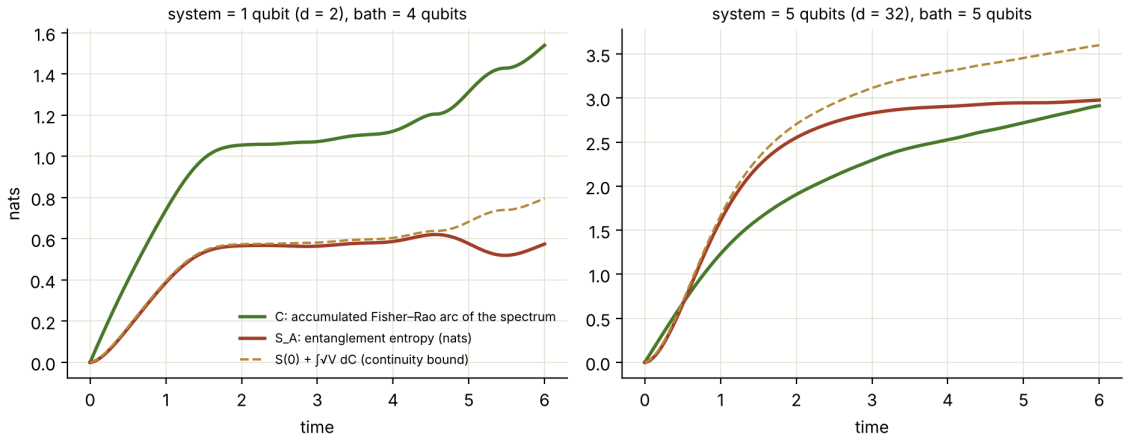


Figure 1. Accumulated Fisher–Rao arc C of the system's spectrum (green), entanglement entropy S_A (red) and the bound of Proposition 2 (gold, dashed), confirmation seed 2000. Left: a one-qubit system, where C bounds S_A. Right: a five-qubit system, where S_A overtakes C while the bound of Proposition 2 holds.

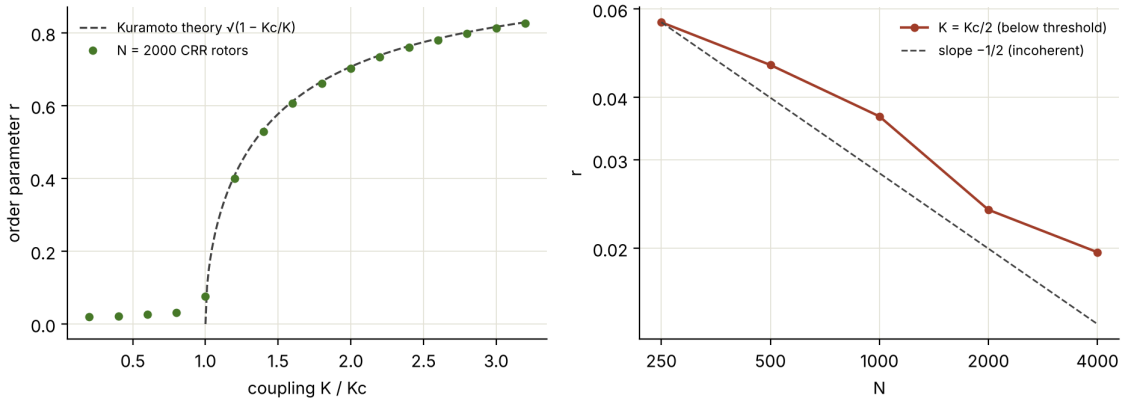


Figure 2. Left: order parameter of N = 2,000 coupled CRR rotors against coupling (green) and the Kuramoto prediction (dashed). Right: below threshold (K = Kc/2), r falls with N toward the incoherent slope −1/2.

5. Discussion: the boundary as a rate

5.1 Inheriting the impossibility

The results leave the impossibility theorem intact, and in two respects they sharpen it:

- **From inside, the agent's reading of its boundary entropy is biased upward in every frame but one** (Proposition 1).
- **The temporal bound gives no access to the forbidden quantity.** Evaluating $\int \sqrt{V}\, dC$ requires the system's own spectrum, which is exactly what it cannot read across its boundary (Proposition 2).

CRR does not offer a way around the unmeasurability. What it changes is the picture of what lies beyond the boundary.

5.2 What lies beyond

The screen account. A definite partition stands between self and environment at every instant, and a determinate far side is kept behind it.

The CRR reading. CRR does not begin with a partition. What plays the role of the boundary is the advancing front of an occasion: its accumulated path toward the next cut. Every system has its own path and dynamics, so there is no single screen, only many rates meeting.

The present. It is the upper limit of the coherence integral and carries no content of its own. Content is what has been accumulated, and the next occasion is seeded only from completed ones.

The far side. It is therefore not a hidden, co-present environment but the settled past of other finite systems, taken up selectively by each system's regeneration. Proposition 2 gives this temporal reading a quantitative form: the correlation that cannot be measured across the boundary is bounded by what has been accumulated along the past.

Epistemic limit and ontological claim. On the screen account, emptiness is an epistemic limit with a determinate far side. On the CRR reading it is an ontological claim: there is no wall, only earlier.

Both statements hold by construction. That the boundary is unevidenceable is what the QRF formalism forbids. That only the past has content is what CRR's definitions entail. Stating both plainly makes the two frameworks comparable rather than rival.

5.3 The collective now

Section 3.4 gives the co-construction of a shared present a precise criterion. Below a critical coupling, finite systems keep private presents, and their shared component vanishes as $N^{-1/2}$. Above it, a common present emerges as a macroscopic order parameter. The transition is the classical Kuramoto transition; calling it a collective now is an interpretation of it.

5.4 Relation to the earlier formulation

An earlier presentation of this argument [13] used a previous formulation of CRR. It differed in four ways:

- the cut was read as Cramér–Rao saturation, $C \cdot \Omega = 1$;
- bistable and rotational systems carried thresholds differing by a factor of two;
- a parameter-free prediction for breathing variability followed from this;
- the regeneration kernel was exponential.

The current formulation replaces these with a single half-turn threshold for every carrier, timing variability set by dynamics, and a normalised regeneration. Two results are refined:

- the self-measurement obstruction, previously illustrated on examples, is now a general proposition;
- the temporal bound on entanglement, previously observed for a single qubit, is now an exact bound whose constant depends on dimension, which explains why the outright inequality $C \geq S_A$ held in that case.

5.5 Interpretive resonances

Several contemplative and process traditions have names for features of this reading. They carry no formal weight here.

- **Nāgārjuna** holds that for one for whom emptiness is possible, everything is possible, and that dependent arising is the middle way (*Mūlamadhyamakakārikā* 24.14, 24.18 [5]). That is emptiness as enabling rather than privative, as a present without content of its own lets every past arrive.
- **Lurianic *tzimtzum*** names a vacated centre [8].
- **Zhuangzi's "pivot of the Way"** names a centre that responds without end because it is fixed nowhere [6].
- **Ibn 'Arabī's "owner of the moment"** names agency over what is taken up from the past rather than over the rate of occasions [7].
- **Doctor and colleagues' care and cognitive light cone** [4] read naturally as the reach of what a system can hold as a goal, and the distance it still has to travel before its next cut.

5.6 Limitations

- **The numerical results are checks of exact statements under chosen models**, not contact with empirical data.
- **Coherence is computed on the system's spectrum rather than its full state.** Eigenvector motion lengthens the arc, so the bound and its failures both persist.
- **The Hamiltonians form one ensemble (GUE).** The rotors are noise-free, with a deterministic sample of tilts.
- **The evidential status is preliminary.** The confirmation was preregistered and anchored, but run without an owner-signed tag or an independent operator. Its evidential ceiling under the Phoenix protocol is PASS-0.

6. Conclusion

Read through Coherence–Rupture–Regeneration, the unevidenceable boundary keeps its impossibility and gains three precise companions:

- an agent's frame-level reading of its own boundary entropy always overshoots;
- the forbidden correlation is bounded by accumulated Fisher–Rao arc weighted by root varentropy, outright for small systems and in weighted form at every scale;
- many finite systems share a present exactly when their coupling exceeds a critical value.

On this account the boundary is not a surface with a hidden far side but a rate: the advancing front of each system's accumulation. Its far side is the settled past of every other finite system, met at a present with no content of its own.

Data and code availability

All code is listed in full in the Appendix and is held in the Phoenix repository:

- the study: `Phoenix_Egg/studies/BOUNDARY-RATE-CHECK-2026-10-27/;`
- the receipts: `Phoenix_Egg/receipts/BOUNDARY-RATE-CHECK-2026-10-27/.`

Requirements: Python 3.13, numpy 2.5.3 and scipy 1.18.1, plus matplotlib for the figures. `sh run.sh RUNS_DIR` reproduces §4 (about 10 minutes), and `python3 figures.py` reproduces Figures 1–2.

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Appendix: complete code

Each file is reproduced verbatim from the frozen study directory.

boundary_model.py

```
        """Model code for BOUNDARY-RATE-CHECK-2026-10-27 (frozen). Rebuilds the formal checks of
        "The Boundary is a Rate" (June 2026) under the current CRR form (CORE/MATHEMATICS.md R1-
R4).

        Three parts, each a known result exercised on fresh random instances:
         1. Self-measurement overshoot (Schur-Horn): the Shannon entropy of measurement outcomes of
rho in any basis
            is >= the von Neumann entropy S(rho), with equality in the eigenbasis.
         2. Retensing as continuity: for a spectrum p(t) moving with Fisher-Rao speed, |dS/ds| <=
sqrt(V(p)), V the
            varentropy (variance of -ln p). Hence S(t) <= S(0) + int sqrt(V) dC, with C the
accumulated Fisher-Rao arc of
            the spectrum (CRR coherence at sigma = 1). sup sqrt(V) is 0.663 nats for a qubit but
exceeds 1 from d = 4,
            so the June claim C >= S_A is a qubit property, not a law.
         3. The collective now: N CRR rotors on circle carriers with the R4 law (no potential, no
noise),
            ds_i/dt = tau_i + K r sin(psi - s_i), tau_i Lorentzian (half-width gamma): the Kuramoto
model.
            r(K) -> sqrt(1 - Kc/K), Kc = 2 gamma (Kuramoto 1975; exact for the Lorentzian).
        """
        import numpy as np
        from numpy.linalg import eigh, eigvalsh

        # ---------- 1. self-measurement overshoot ----------
        def haar_unitary(d, rng):
            z = (rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d))) / np.sqrt(2)
            q, r = np.linalg.qr(z); ph = np.diag(r) / np.abs(np.diag(r)); return q * ph

        def random_state(d, rng):
            g = rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d)); rho = g @
g.conj().T; return rho / np.trace(rho).real

        def H(p):
            p = np.clip(np.real(p), 0, None); p = p / p.sum(); nz = p > 0; return float(-(p[nz] *
np.log(p[nz])).sum())

        def overshoot(d, rng):
            rho = random_state(d, rng); U = haar_unitary(d, rng)
            S = H(eigvalsh(rho)); Hb = H(np.diag(U.conj().T @ rho @ U)); w, V = eigh(rho); He =
H(np.diag(V.conj().T @ rho @ V))
            return Hb - S, abs(He - S)

        # ---------- 2. retensing ----------
        def gue(d, rng):
            A = rng.standard_normal((d, d)) + 1j * rng.standard_normal((d, d)); return (A +
A.conj().T) / (2 * np.sqrt(2 * d))

        def trajectory(nA, nB, seed, g=1.0, tmax=6.0, npts=241):
            """System of nA qubits, bath of nB qubits, H = H_A + H_B + g H_AB (all GUE), product
initial state."""
            rng = np.random.default_rng(seed); dA, dB = 2 ** nA, 2 ** nB
            Hm = np.kron(gue(dA, rng), np.eye(dB)) + np.kron(np.eye(dA), gue(dB, rng)) + g * gue(dA
* dB, rng)
            E, V = eigh(Hm)
            a = rng.standard_normal(dA) + 1j * rng.standard_normal(dA); a /= np.linalg.norm(a)
            b = rng.standard_normal(dB) + 1j * rng.standard_normal(dB); b /= np.linalg.norm(b)
            c0 = V.conj().T @ np.kron(a, b); ts = np.linspace(0, tmax, npts); P = []; norm_err =
0.0
```

```
            for t in ts:
                psi = V @ (np.exp(-1j * E * t) * c0); norm_err = max(norm_err, abs(np.vdot(psi,
psi).real - 1))
                M = psi.reshape(dA, dB); ev = np.clip(eigvalsh(M @ M.conj().T).real, 1e-300, None);
P.append(np.sort(ev / ev.sum())[::-1])
            P = np.array(P); S = -(P * np.log(P)).sum(1)
            step = 2 * np.arccos(np.clip(np.sqrt(P[1:] * P[:-1]).sum(1), -1, 1))
            C = np.concatenate([[0.0], np.cumsum(step)])
            l = np.log(P); m = (P * l).sum(1, keepdims=True); sV = np.sqrt((P * (l - m) **
2).sum(1))
            # sup of sqrt(V) along each chord, taken at the endpoints plus a midpoint spectrum
(conservative for small steps)
            mid = 0.5 * (P[1:] + P[:-1]); lm = np.log(mid); mm = (mid * lm).sum(1, keepdims=True);
sVm = np.sqrt((mid * (lm - mm) ** 2).sum(1))
            sup = np.maximum(np.maximum(sV[1:], sV[:-1]), sVm)
            bound = S[0] + np.concatenate([[0.0], np.cumsum(step * sup)])
            return dict(ts=ts, S=S, C=C, bound=bound, norm_err=norm_err)

        def max_root_varentropy(d, starts=60, seed=0):
            from scipy.optimize import minimize
            rng = np.random.default_rng(seed); best = 0.0
            def f(x):
                p = np.exp(x - x.max()); p /= p.sum(); l = np.log(np.clip(p, 1e-300, None)); m = (p
* l).sum(); return -np.sqrt((p * (l - m) ** 2).sum())
            for _ in range(starts):
                r = minimize(f, rng.normal(size=d) * 3, method="Nelder-Mead",
options=dict(maxiter=20000, xatol=1e-10, fatol=1e-12)); best = max(best, -r.fun)
            return best

        def geodesic_counterexample(d):
            """Fisher-Rao geodesic length from a pure spectrum to the uniform one vs the entropy
gained (ln d)."""
            return 2 * np.arccos(1 / np.sqrt(d)), np.log(d)

        # ---------- 3. collective now ----------
        def kuramoto_r(N, K, gamma, seed, T=300.0, dt=0.05, burn=150.0):
            """Mean order parameter of N CRR rotors (R4 law, no potential/noise) with deterministic
Lorentzian tilts."""
            rng = np.random.default_rng(seed)
            q = (np.arange(1, N + 1) / (N + 1)) - 0.5; tau = gamma * np.tan(np.pi * q)
            s = rng.uniform(0, 2 * np.pi, N); rs = []; nsteps = int(T / dt); nb = int(burn / dt)
            for k in range(nsteps):
                z = np.exp(1j * s).mean(); r, psi = abs(z), np.angle(z)
                s = s + dt * (tau + K * r * np.sin(psi - s))
                if k >= nb: rs.append(r)
            return float(np.mean(rs))
```

run_checks.py

```
        """Holdout runner for BOUNDARY-RATE-CHECK-2026-10-27. Usage: python3 -I run_checks.py
OUT.json
        Holdout seeds: overshoot draws from default_rng(2000); trajectories seeds 2000-2005;
Kuramoto seeds 2000-2002."""
        import json, sys, pathlib
        sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent))
        import numpy as np
        from boundary_model import overshoot, trajectory, kuramoto_r, max_root_varentropy,
geodesic_counterexample

        def rnd(x):
            if isinstance(x, float): return round(x, 6)
            if isinstance(x, list): return [rnd(v) for v in x]
            if isinstance(x, dict): return {k: rnd(v) for k, v in x.items()}
            return x

        out = {}
        rng = np.random.default_rng(2000)
```

```
        out["overshoot"] = {}
        for d in (2, 4, 8, 16):
            o = [overshoot(d, rng) for _ in range(500)]
            out["overshoot"][str(d)] = dict(n=500, min_gap=float(min(x[0] for x in o)),
mean_gap=float(np.mean([x[0] for x in o])),
                                            max_eigenbasis_err=float(max(x[1] for x in o)))
        out["retensing"] = {}
        for nA, nB in [(1, 4), (2, 4), (3, 5), (5, 5)]:
            rows = []
            for seed in range(2000, 2006):
                r = trajectory(nA, nB, seed)
                rows.append(dict(seed=seed, min_C_minus_S=float(np.min(r["C"] - r["S"])),
min_bound_minus_S=float(np.min(r["bound"] - r["S"])),
                                 max_S_over_C=float(np.max(r["S"][1:] / r["C"][1:])),
final_S=float(r["S"][-1]), final_C=float(r["C"][-1]), norm_err=float(r["norm_err"])))
            out["retensing"][str(2 ** nA)] = rows
        out["kappa"] = {str(d): max_root_varentropy(d) for d in (2, 4, 8, 32)}
        out["geodesic"] = {str(d): dict(zip(("C_geodesic", "S_uniform"), map(float,
geodesic_counterexample(d)))) for d in (13, 14, 32, 64)}
        g = 0.5; Kc = 2 * g
        out["kuramoto_above"] = {str(m): dict(theory=float(np.sqrt(1 - 1 / m)), r=[kuramoto_r(4000,
m * Kc, g, s) for s in range(2000, 2003)]) for m in (1.5, 2.0, 3.0)}
        Ns = [250, 500, 1000, 2000, 4000]
        rs = [float(np.mean([kuramoto_r(N, 0.5 * Kc, g, s) for s in range(2000, 2003)])) for N in
Ns]
        out["kuramoto_below"] = dict(N=Ns, r=rs, slope=float(np.polyfit(np.log(Ns), np.log(rs), 1)
[0]))
        pathlib.Path(sys.argv[1]).write_text(json.dumps(rnd(out), indent=1, sort_keys=True) + "\n")
```

score.py

```
        """Frozen scorer for BOUNDARY-RATE-CHECK-2026-10-27. Usage: python3 -I score.py
outcomes.json > verdict.json"""
        import json, sys
        o = json.load(open(sys.argv[1])); R = o["retensing"]
        checks = {
         "V0_harness_unitarity": all(r["norm_err"] < 1e-10 for d in R for r in R[d]),
         "B1_overshoot_schur_horn": all(v["min_gap"] >= -1e-10 and v["mean_gap"] > 0 and
v["max_eigenbasis_err"] < 1e-10 for v in o["overshoot"].values()),
         "B2_continuity_bound_holds": all(r["min_bound_minus_S"] >= -1e-9 for d in R for r in
R[d]),
         "B3_qubit_C_bounds_S": all(r["min_C_minus_S"] >= -1e-9 for r in R["2"]),
         "B4_C_ge_S_fails_at_dA_32": all(r["min_C_minus_S"] < -0.3 for r in R["32"]),
         "B5_kappa_crosses_one": o["kappa"]["2"] < 1 < o["kappa"]["4"],
         "B6_geodesic_counterexample": o["geodesic"]["14"]["C_geodesic"] < o["geodesic"]["14"]
["S_uniform"] and o["geodesic"]["13"]["C_geodesic"] > o["geodesic"]["13"]["S_uniform"],
         "B7_kuramoto_above_Kc": all(abs(sum(v["r"]) / len(v["r"]) - v["theory"]) <= 0.03 for v in
o["kuramoto_above"].values()),
         "B8_kuramoto_below_Kc_finite_size": -0.65 <= o["kuramoto_below"]["slope"] <= -0.30,
        }
        verdict = "VOID" if not checks["V0_harness_unitarity"] else ("RESULT" if
all(checks.values()) else "FAIL")
        json.dump({"study_id": "BOUNDARY-RATE-CHECK-2026-10-27", "checks": checks, "verdict":
verdict}, sys.stdout, indent=1, sort_keys=True); print()
```

controls.py

```
        """Controls for BOUNDARY-RATE-CHECK-2026-10-27, scored before the freeze.
        pos_bound_is_tight: along an entropy-gradient flow of the spectrum (d = 8) the continuity
bound is nearly attained,
          so it is not vacuous (ratio of entropy gained to int sqrt(V) dC >= 0.99).
        neg_geodesic_breaks_C_ge_S: along the Fisher-Rao geodesic from a pure to the uniform
spectrum at d = 32, C < S at the end
          (the scorer's B4 logic must be able to register a violation).
        null_kuramoto_K0: with no coupling r ~ 0.886/sqrt(N) (incoherent), r*sqrt(N) in [0.5, 1.5]
```

```
at N = 2000.
        Usage: python3 -I controls.py > controls.json"""
        import json, sys, pathlib
        sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent))
        import numpy as np
        from boundary_model import kuramoto_r
        d = 8; p = np.array([0.93] + [0.01] * 7); S0 = -(p * np.log(p)).sum(); integral = 0.0
        for _ in range(20000):                      # gradient flow of S in the Fisher metric:
dp_i = eps p_i (-ln p_i - S)
            l = np.log(p); S = -(p * l).sum(); dp = 1e-4 * p * (-l - S); q = p + dp; q /= q.sum()
            ds = 2 * np.arccos(min(1.0, np.sqrt(p * q).sum())); lm = np.log(p); m = (p * lm).sum();
integral += ds * np.sqrt((p * (lm - m) ** 2).sum()); p = q
        gain = -(p * np.log(p)).sum() - S0
        x = np.linspace(0, 1, 2001); d2 = 32; e = np.zeros(d2); e[0] = 1; u = np.ones(d2) / d2   #
geodesic on the sphere of sqrt-probabilities
        th = np.arccos(np.sqrt(e * u).sum()); path = [((np.sin((1 - t) * th) * np.sqrt(e) +
np.sin(t * th) * np.sqrt(u)) / np.sin(th)) ** 2 for t in x]
        C_end = sum(2 * np.arccos(min(1.0, np.sqrt(a * b).sum())) for a, b in zip(path[:-1],
path[1:])); S_end = np.log(d2)
        r0 = kuramoto_r(2000, 0.0, 0.5, 7000)
        out = {"pos_bound_is_tight": {"ratio": round(gain / integral, 4), "pass": bool(gain /
integral >= 0.99)},
               "neg_geodesic_breaks_C_ge_S": {"C_end": round(C_end, 4), "S_end": round(S_end, 4),
"violation_registered": bool(C_end < S_end)},
               "null_kuramoto_K0": {"r_sqrtN": round(r0 * np.sqrt(2000), 3), "pass": bool(0.5 <= r0
* np.sqrt(2000) <= 1.5)}}
        out["controls_behave"] = out["pos_bound_is_tight"]["pass"] and
out["neg_geodesic_breaks_C_ge_S"]["violation_registered"] and out["null_kuramoto_K0"]["pass"]
        print(json.dumps(out, indent=1, sort_keys=True))
```

run.sh

```
        #!/bin/sh
        # usage: sh run.sh RUNS_DIR   (after the freeze and anchor; holdout seeds 2000+ are drawn
for the first time)
        set -e
        H="$(cd "$(dirname "$0")" && pwd)"; R="$1"; mkdir -p "$R"
        export PYTHONDONTWRITEBYTECODE=1
        python3 -I "$H/run_checks.py" "$R/outcomes.json"
        python3 -I "$H/score.py" "$R/outcomes.json" > "$R/verdict.json"
```

figures.py

```
        """Figures for the rebuilt "The Boundary is a Rate" (holdout seed 2000 / Kuramoto seed
2000, identical to the frozen run).
        Usage: python3 figures.py   (writes ../fig_retensing.png and ../fig_collective.png)"""
        import numpy as np
        import matplotlib; matplotlib.use("Agg")
        import matplotlib.pyplot as plt
        from boundary_model import trajectory, kuramoto_r

        G, GOLD, RUST, INK = "#4A7A2C", "#B8893A", "#A3402B", "#444444"
        fig, axes = plt.subplots(1, 2, figsize=(9.6, 3.9), dpi=200)
        for ax, (nA, nB, lab) in zip(axes, [(1, 4, "system = 1 qubit (d = 2), bath = 4 qubits"),
(5, 5, "system = 5 qubits (d = 32), bath = 5 qubits")]):
            r = trajectory(nA, nB, 2000)
            ax.plot(r["ts"], r["C"], color=G, lw=2, label="C: accumulated Fisher–Rao arc of the
spectrum")
            ax.plot(r["ts"], r["S"], color=RUST, lw=2, label="S_A: entanglement entropy (nats)")
            ax.plot(r["ts"], r["bound"], color=GOLD, lw=1.4, ls="--", label="S(0) + ∫√V dC
(continuity bound)")
            ax.set_title(lab, fontsize=9.5); ax.set_xlabel("time")
            for sp in ("top", "right"): ax.spines[sp].set_visible(False)
            ax.grid(color="#e6e2d8", lw=0.6)
        axes[0].set_ylabel("nats"); axes[0].legend(frameon=False, fontsize=7.5, loc="lower right")
```

```
        fig.tight_layout(); fig.savefig("../fig_retensing.png", facecolor="white")

        fig, (a1, a2) = plt.subplots(1, 2, figsize=(9.6, 3.6), dpi=200); g = 0.5; Kc = 2 * g
        Ks = np.linspace(0.2, 3.2, 16) * Kc; rs = [kuramoto_r(2000, K, g, 2000) for K in Ks]
        kk = np.linspace(Kc, 3.2 * Kc, 200)
        a1.plot(kk / Kc, np.sqrt(1 - Kc / kk), color=INK, lw=1.2, ls="--", label="Kuramoto theory
√(1 − Kc/K)")
        a1.plot(Ks / Kc, rs, "o", color=G, ms=4, label="N = 2000 CRR rotors")
        a1.set_xlabel("coupling K / Kc"); a1.set_ylabel("order parameter r");
a1.legend(frameon=False, fontsize=8)
        Ns = [250, 500, 1000, 2000, 4000]; rb = [kuramoto_r(N, 0.5 * Kc, g, 2000) for N in Ns]
        a2.loglog(Ns, rb, "o-", color=RUST, ms=4, label="K = Kc/2 (below threshold)")
        a2.loglog(Ns, rb[0] * (np.array(Ns) / Ns[0]) ** -0.5, color=INK, lw=1, ls="--",
label="slope −1/2 (incoherent)")
        a2.minorticks_off(); a2.set_xticks(Ns); a2.set_xticklabels([str(n) for n in Ns]);
a2.set_yticks([0.02, 0.03, 0.04, 0.06]); a2.set_yticklabels(["0.02", "0.03", "0.04", "0.06"])
        a2.set_xlabel("N"); a2.set_ylabel("r"); a2.legend(frameon=False, fontsize=8)
        for ax in (a1, a2):
            for sp in ("top", "right"): ax.spines[sp].set_visible(False)
            ax.grid(color="#e6e2d8", lw=0.6)
        fig.tight_layout(); fig.savefig("../fig_collective.png", facecolor="white")
```